This equation is based on the assumption that the compound nucleus is the center of reference. It should be noted that the height of the Coulomb barrier does not influence the excitation energy of the nucleus in any way other than that a projectile must have a kinetic energy greater than that value before the reaction can occur. In the reaction all E_{kin} and E_{coul} are transferred into E_{kin}, E_{exc} and E^0_{mass} for the compound nucleus. The excitation of the compound nucleus can be removed either through the emission of γ-rays, neutrons, or by the decay of the compound nucleus into different product nuclei. In the former case, $E_{kin} = E_\upsilon$. In the latter case we again have to bring in (10.10).

If the reaction passes through formation of a compound nucleus, having a lifetime of many nuclear vibrations, we should expect that the amounts and types of reaction products were independent of the projectile − target combination used to produce a specific compound nucleus and only depend on its excitation energy. Essentially the compound nuclear does not have any memory of how it was formed and its decay is controlled by its own features only. This has indeed been found to be the case for some kinds of nuclear reactions, see Ch. 11.

10.9. RADIOACTIVE NEUTRON SOURCES

Neutrons were discovered through the reaction between α-particles, emitted by radioactive substances, and light elements: ^{4}He (from Po) + ^{9}Be → ^{12}C + n (Chadwick 1932). The coulomb barrier of light elements is sufficiently small to be penetrated by α-particles emitted by radioactive nuclides; in the reaction used by Chadwick the coulomb barrier is only ∼3.5 MeV, eqn. (10.15) and Fig. 10.3. All early neutron research was conducted with sources of this kind, the most popular being the RaBe mixture. A number of radioactive n-sources are listed in Table 10.2.

Table 10.2 Radioactive neutron sources

Material	Half-life	Neutron yield (n s^{-1} Bq^{-1})	γ-dose rate[(a)] (mSv h^{-1} m^2 Bq^{-1})
^{226}Ra + Be	1600 y	3.5×10^{-4}	850
^{239}Pu + Be	24110 y	2.4×10^{-4}	4
^{239}Pu + ^{18}O	24110 y	7.8×10^{-6}	[(b)]
^{241}Am + Be	433 y	6.8×10^{-5}	2.5
^{210}Po + Be	138 d	6.8×10^{-5}	<0.3[(c)]
^{124}Sb + Be	60 d	$\sim5\times10^{-6}$	1000
^{252}Cf	2.6 y	5×10^{12} n s^{-1} g^{-1}	[(d)]

[(a)] Dose rate at 1 m from source, cf. eqn. (7.9).
[(b)] Variable source of $Pu^{16}O_2 + H_2$ ^{18}O; by transferring water through heating into the porous oxide, the reaction ^{18}O(α,n)^{21}Ne is initiated; by cooling the water evaporates and is returned to storage whereby the n output decreases to < 0.01 of maximum output.
[(c)] The half-life can be increased to 22 y by including the precursor nuclide ^{210}Pb.
[(d)] γ-dose rate increases with time due to formation of γ-emitting daughters.

The nuclide $^{252}_{98}$Cf emits neutrons through spontaneous fission in ~3% of all decays, the rest being α-decays. All the other neutron sources listed involve a radioactive nuclide whose decay causes a nuclear reaction in a secondary substance which produces neutrons. For example, $^{124}_{51}$Sb produces neutrons in beryllium powder or metal as a result of the initial emission of γ-rays, in which case there is no coulomb barrier to penetrate. Radium, polonium, plutonium, and americium produce neutrons by nuclear reactions induced in beryllium by the α-particles from their radioactive decay. For the neutrons produced either by spontaneous fission in californium or by the α-particle reaction with beryllium, the energy is between 0 and 10 MeV, while for the neutrons emitted in the γ,n-reaction involving $^{124}_{51}$Sb, the energy is 0.02 MeV. Neutron sources are commercially available.

Neutron multipliers have also come into use. These usually consist of a tank containing highly enriched ^{235}U. The uranium releases 100 n for each trigger neutron sent in; the common source is ^{252}Cf. These units are used for neutron radiography (§7.9.2), activation analysis (§18.3.3), etc.

Although the neutron flux from radioactive sources is comparatively small, they are still quite useful for specific purposes, due to their extreme simplicity, reliability (they cannot be shut off), and small size. The γ-radiation which is also present is, however, a disadvantage. From this point of view, ^{252}Cf is the most preferable neutron source, but, on the other hand, it has a relatively short half-life compared with some of the other sources. ^{252}Cf is further discussed in Chapter 14.

10.10. EXERCISES

10.1. (a) What kinetic energy must be given to a helium atom in order to increase its mass by 1%? (b) What are the mean velocity and the mean kinetic energy of a helium atom at STP?

10.2. Calculate the distance of closest approach for 5 MeV α-particles to a gold target.

10.3. In a Rutherford scattering experiment ^{2}H atoms of 150 keV are used to bombard a thin ^{58}Ni foil having a surface density of 67×10^{-6} g cm^{-2}. The detector subtends a solid angle of 1.12×10^{-4} sr and detects 4816 deuterons out of a total of 1.88×10^{12} incident on target. Calculate (a) the differential cross-section (in barns). (b) What is the distance between target and the solid state detector, which has a surface area of 0.2 cm^2?

10.4. In Rutherford scattering on a silver foil using α-particles from a thin-walled radon tube, the following data were observed: $d\sigma/d\Omega = 22(\theta\ 150°)$, 47(105°), 320(60°), 5260(30°), 105 400(15°) barns per steradian. Calculate the energy of the incident α-particles.

10.5. Alpha-particles from ^{218}Po (E_α 6.0 MeV) are used to bombard a gold foil. (a) How close to the gold nucleus can these particles reach? (b) What is the nuclear radius of gold according to the radius-mass relation ($r_o = 1.3$ fm)?